Library Crowd Monitoring

& Smart Lighting System

ESP32 · IR Sensors · LCD I2C · Relay Module

Embedded Systems | IoT | Smart Library

Project Overview

What is it?

A real-time system that monitors the number of people inside a library using IR sensors at entry and exit points.

An ESP32 microcontroller processes the signals, maintains a live occupancy count, and displays the crowd status on a 16×2 LCD screen.

Library visitors and staff can instantly see whether the space is Empty, Medium, or Crowded.

2

IR Sensors

3

Status Levels

Real

Time Monitoring

Objectives

01 Monitor library occupancy in real time

02 Display Empty / Medium / Crowded status on LCD

03 Help users assess crowd level before entering

04 Automate lighting based on live occupancy count

05 Deliver a low-cost embedded-systems solution

06 Reduce overcrowding in study spaces

System Architecture

INPUT	PROCESSING	OUTPUT
IR Entry Sensor (GPIO 27)	ESP32 Microcontroller	LCD 16×2 (I2C Display)
IR Exit Sensor (GPIO 25)	Real-time Counter Logic	Relay Module (Lighting)

Smart Lighting Feature

How It Works

1. IR sensors detect entry / exit
2. ESP32 updates count in real time
3. count > 0 → GPIO 26 LOW → Relay closes → Lights ON
4. count = 0 → GPIO 26 HIGH → Relay opens → Lights OFF

Why We Added It

Energy Saving
Lights only ON when occupied

Fully Automatic
No switches, no manual steps

Eco-Friendly
Reduced electricity waste

Better UX
Always lit when someone enters

System Components

ESP32

Microcontroller

`Pin: Main brain`

Reads sensors, updates LCD & relay

IR Sensor Entry

People Counting

`Pin: GPIO 27`

Detects beam break at entrance

IR Sensor Exit

People Counting

`Pin: GPIO 25`

Detects beam break at exit

LCD 16×2 I2C

Display

`Pin: SDA 33 / SCL 32`

Shows count & crowd status

Relay Module

Lighting Control

`Pin: GPIO 26`

Switches lights ON / OFF

Li-Ion Battery

Power Source

`Pin: 3.3V stable`

Rechargeable, reliable supply

Occupancy Status & Lighting Behaviour

Count	Status	Lighting	Description
0	Empty	OFF	No occupants — lights cut automatically
1–3	Empty	ON	Very low occupancy — lights activate
4–7	Medium	ON	Moderate occupancy — library in use
8+	Crowded	ON	High occupancy — library at full capacity

Future Improvements

01

Card-Based Entry

Track user stay duration and predict seat availability

02

Buzzer Alert

Warn occupants before lights turn off automatically

03

Multiple Entrances

Expand coverage to corridors and multiple doors

04

Mobile App

Monitor occupancy remotely from any smartphone

05

Wi-Fi / IoT

Real-time online updates via cloud IoT platform

06

Better Sensor Array

Multiple sensors to improve counting accuracy

A practical, real-time solution for smart library management.

- IR-based people counting gives accurate, live occupancy data
- Automated lighting eliminates energy waste with zero manual effort
- Clear LCD display keeps visitors and staff always informed
- Low-cost hardware makes the solution practical and scalable